Browser Internals

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https://i.imgur.com/dCrq7K7.jpg

Background: Overall Outline

- ~~What constitutes a webpage?~~
- **What goes on inside a Browser?**
  - **What standard security mechanisms implemented?**
- How do client and server communicate?
  - HTTP/HTTPs protocol
  - Session Management via cookies and tokens
- How does a web server process requests and generate responses?
  - Static vs Dynamic content

Outline

- Composition of a Browser (processes within)
- High Level overview of navigation
- Internals of Renderer Process
- Security Mechanisms within Browser

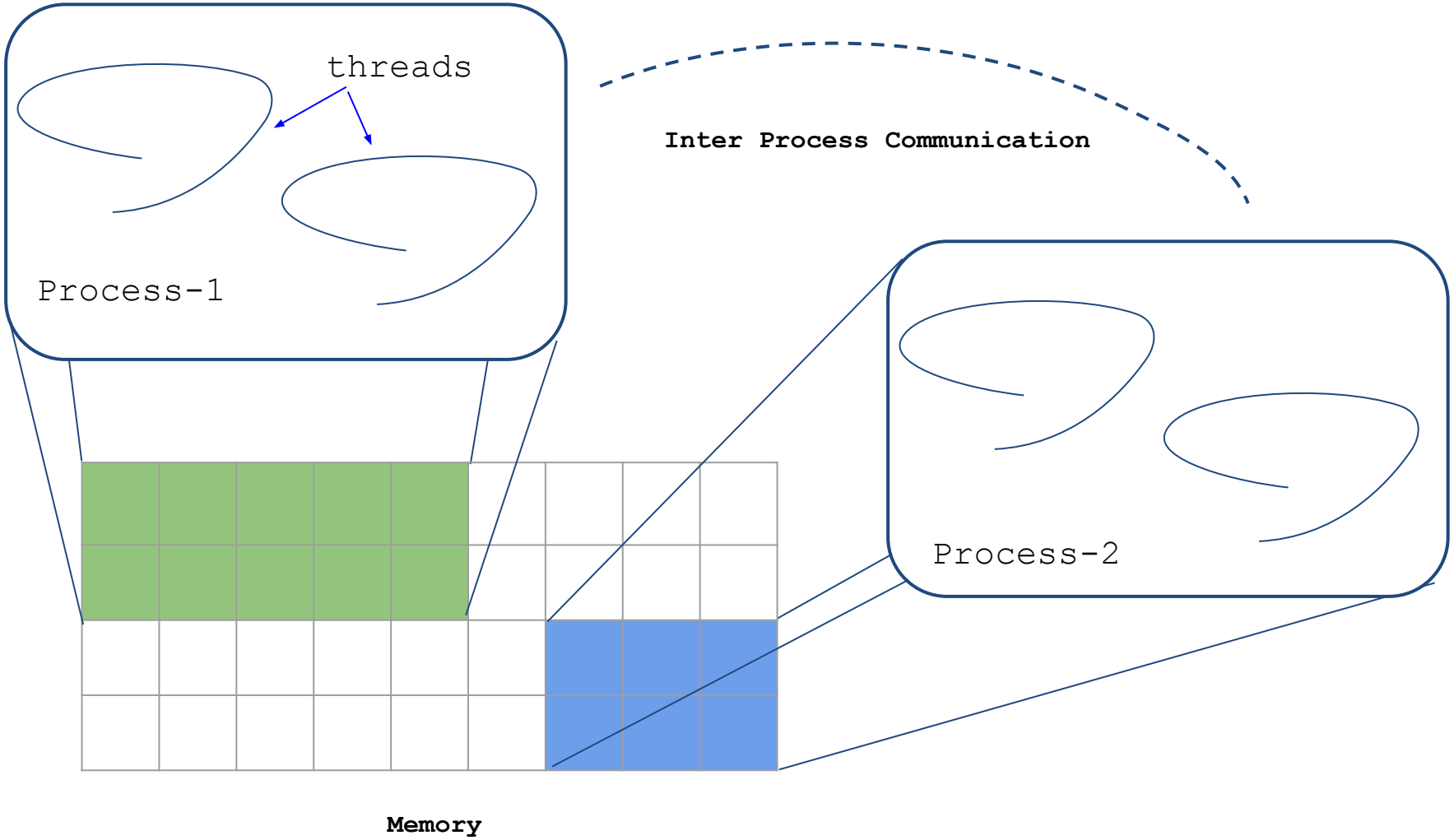
Browser

- A software application that helps users to “browse” the Web
  - Provide a rich user interface
    - Address bar, navigation buttons, bookmarks etc
  - Support extensions or add-ons to enhance functionality
- Popular web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, and Opera

Background: Processes and Threads

- Process: a running program
- Thread: lives inside a process and executes part of the program
- When a process starts, OS gives it a chunk of memory
- Close application, processes stopped and memory freed

- Process can fork other processes
  - These worker processes run different tasks of an application
    - They talk via Inter Process Communication (IPC)
    - Each process has its own “disjoint” chunk of memory
  - One worker unresponsive, can be restarted independent of others → better fault tolerance

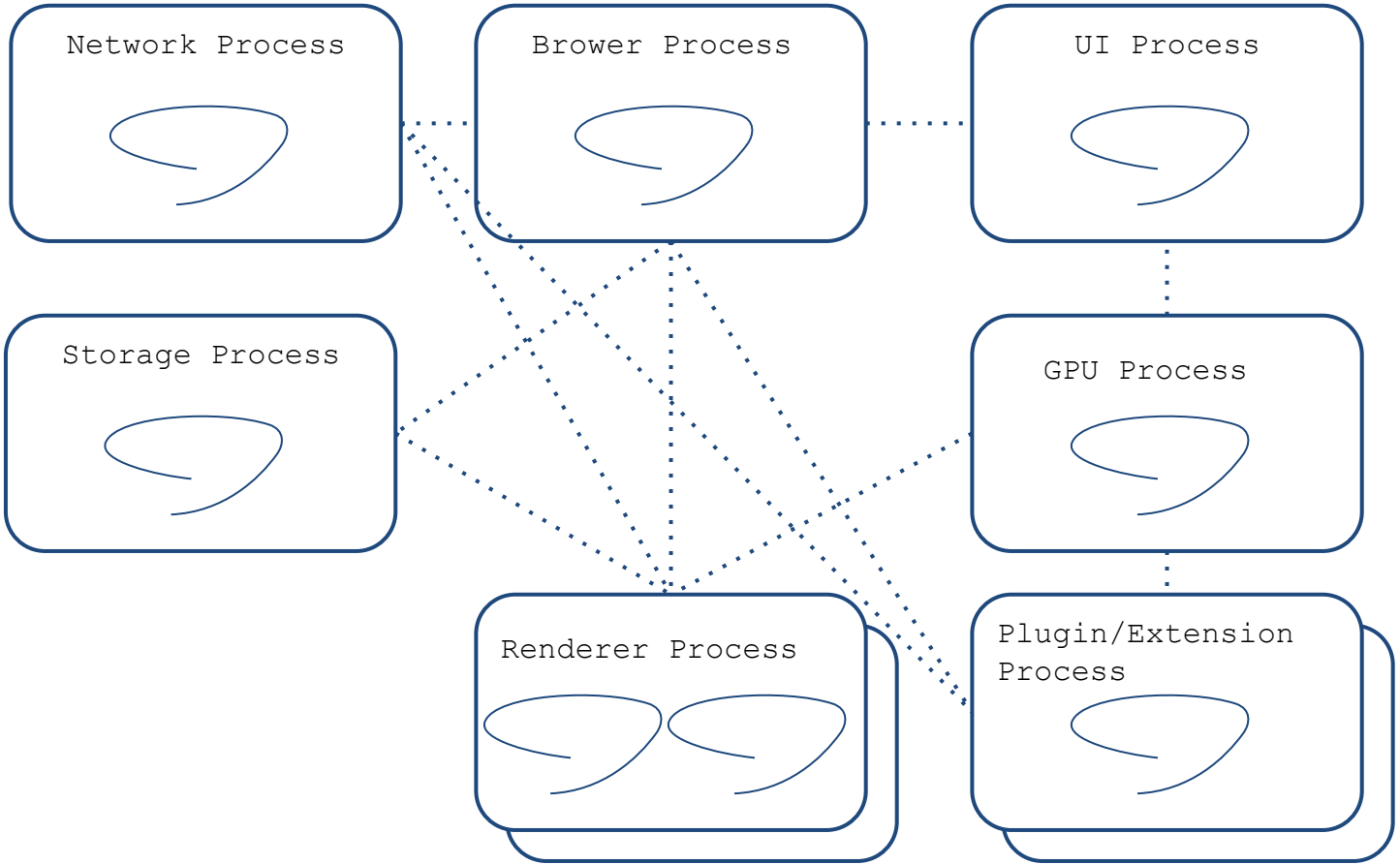

threads
Process-1
Inter Process Communication
Process-2
Memory

Case Study: Chrome browser

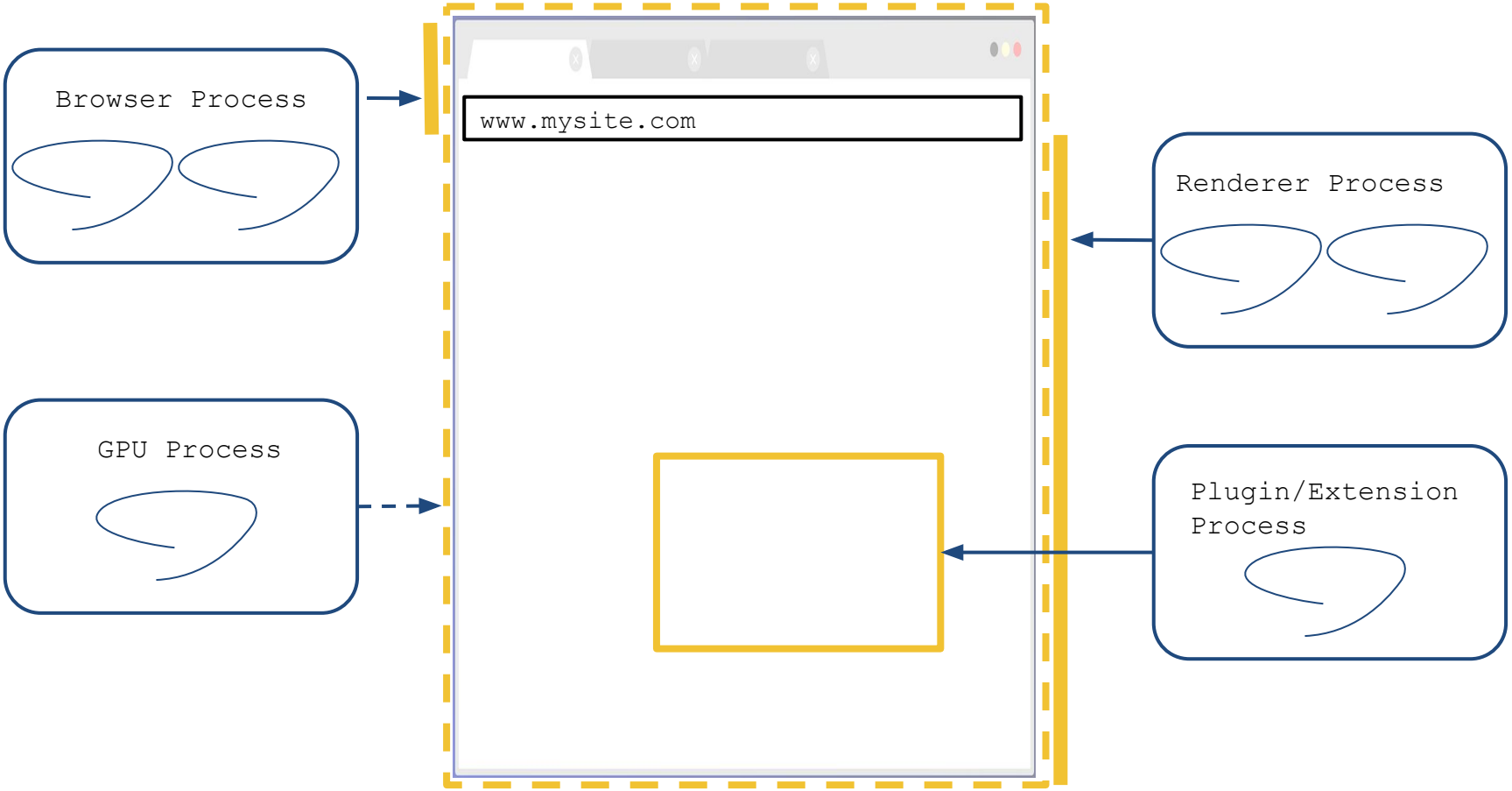
- Start a browser (application)
  - One process with many different threads or many different processes with a few threads?
- How many processes are running?
  - Open Chrome, click three dots →More Tools (top right corner)
  - Select Task Manager
  - Shows a list of processes currently running and their CPU/Memory usage

Task Manager - Google Chrome

Task	Memory footprint	CPU	Network	Process ID
Browser	81,860K	0.0	0	26000
GPU Process	260,128K	1.6	0	23248
Utility: Network Service	20,556K	0.0	0	7660
Utility: Storage Service	15,272K	0.0	0	4148
Utility: Data Decoder Service	16,012K	0.0	0	4700
Renderer	–	0.0	0	15916
Renderer	–	0.0	0	8492
Spare Renderer	20,568K	0.0	0	7652
Tab: Indian Institute of Technology Bombay | IIT Bombay	61,464K	0.0	0	26216
Service Worker: chrome-extension://fheoggkfdfchfphceeifdbe...	36,244K	0.0	0	13796


Network Process
Brower Process
UI Process
Storage Process
GPU Process
Renderer Process
Plugin/Extension Process

Process	Role
Browser	● Controls address bar, bookmarks, back and forward buttons etc ● Underhood, handles privileged stuff such as network requests and file access
Renderer	Controls what happens inside a tab (one tab per website)
Plugin/Extensions	Controls any plugins/extensions (one process per plugin or extension)
GPU	● Responsible for rendering and displaying graphics, images, and multimedia content ● Also handles compositing (final image that you see on your screen)
Storage	Manages different types of storage (cookies, local storage, indexed dB etc) with isolation where appropriate

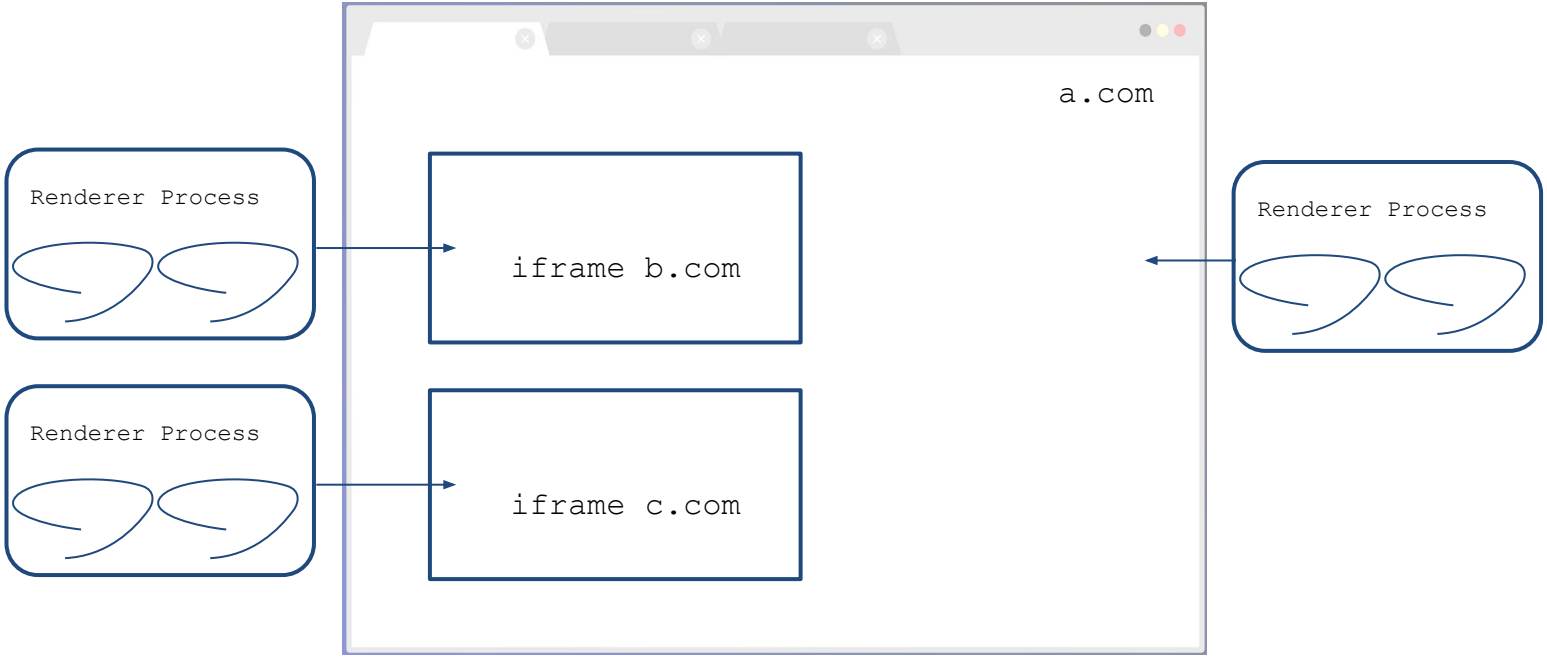

Browser Process
www.mysite.com
Renderer Process
GPU Process
Plugin/Extension Process

A few points to note

- Browser Process:
  - Running on powerful hardware → Splits into different processes giving more stability
  - Less powerful hardware → Runs as single process saving memory
- Renderer Process:
  - One process per tab. Why?
    - If one tab unresponsive, other tabs still active
    - More important reason: Security
      - OS protects one process from accessing another process memory

- iframes and Renderer Process
  - iframe (inline frame) allows embedding content from another website (say B) inside a website (say A)
    - <iframe src="https://www.example.com" width="600" height="400" frameborder="0"></iframe>
      - src attribute specifies the URL of the external site to be displayed in the iframe
  - How many renderer processes run for this website tab of A?

- Each cross-site iframe in same tab still gets a separate renderer process
  - Not easy to implement
  - E.g. Ctrl+F to find a word in a page → searching across different renderer processes

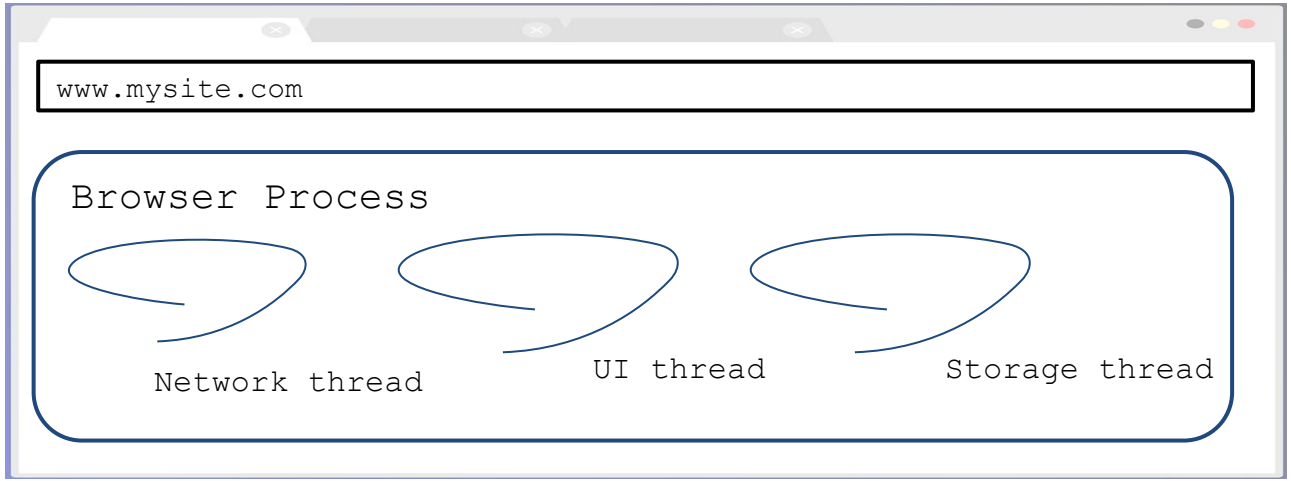


Outline

- ~~Composition of a Browser (processes within)~~
- **High Level overview of navigation**
- Internals of Renderer Process
- Security Mechanisms within Browser

Navigation

- Starts with a browser process
- Browser process runs other threads/processes like UI, network, and storage etc to handle navigation



Steps Involved

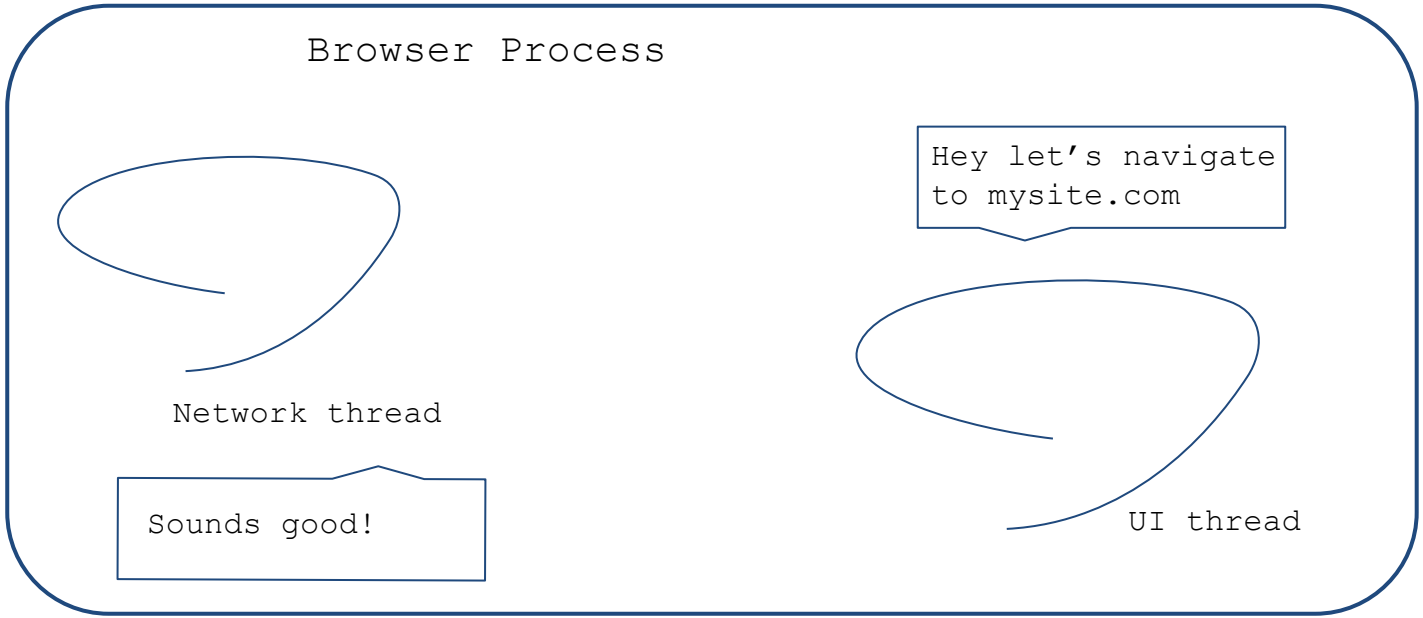
Step 1: User types URL in the address bar

- Handled by the UI thread
  - "Is this a search query or URL?"
  - If search → search engine is invoked
  - If not, initiates a network call to get site content; displays loading spinner

Step 2: Start navigation

- Network thread uses DNS/TLS/TCP to establish a connection with web server
- Server redirect (e.g. 301 status code)? Contact UI thread and another URL request initiated

www.mysite.com



Step3: Read HTTP response

- Network thread looks at the first few bytes of the response header
- Content-Type header indicates type of data
  - If missing MIME Type, sniffing is done
- If response is HTML file, renderer process is invoked via UI thread
- If zip file or some other file → a download request → storage thread invoked via UI thread
- Many safebrowsing checks also happens in network thread
  - Domain matches malicious site → display a warning page!
  - Sensitive cross-site data does not make it to the renderer process (will cover under CORS later)

Step4: Find a Renderer Process

- If ready to navigate to requested site, network thread tells UI thread data is ready
- UI thread finds a renderer process to render the web page
- Speed-up optimization: While UI thread sends URL request to network thread (in step1), can proactively find or start a renderer process
  - If all goes well, renderer process ready in standby

www.mysite.com

Browser Process

I got what you requested!

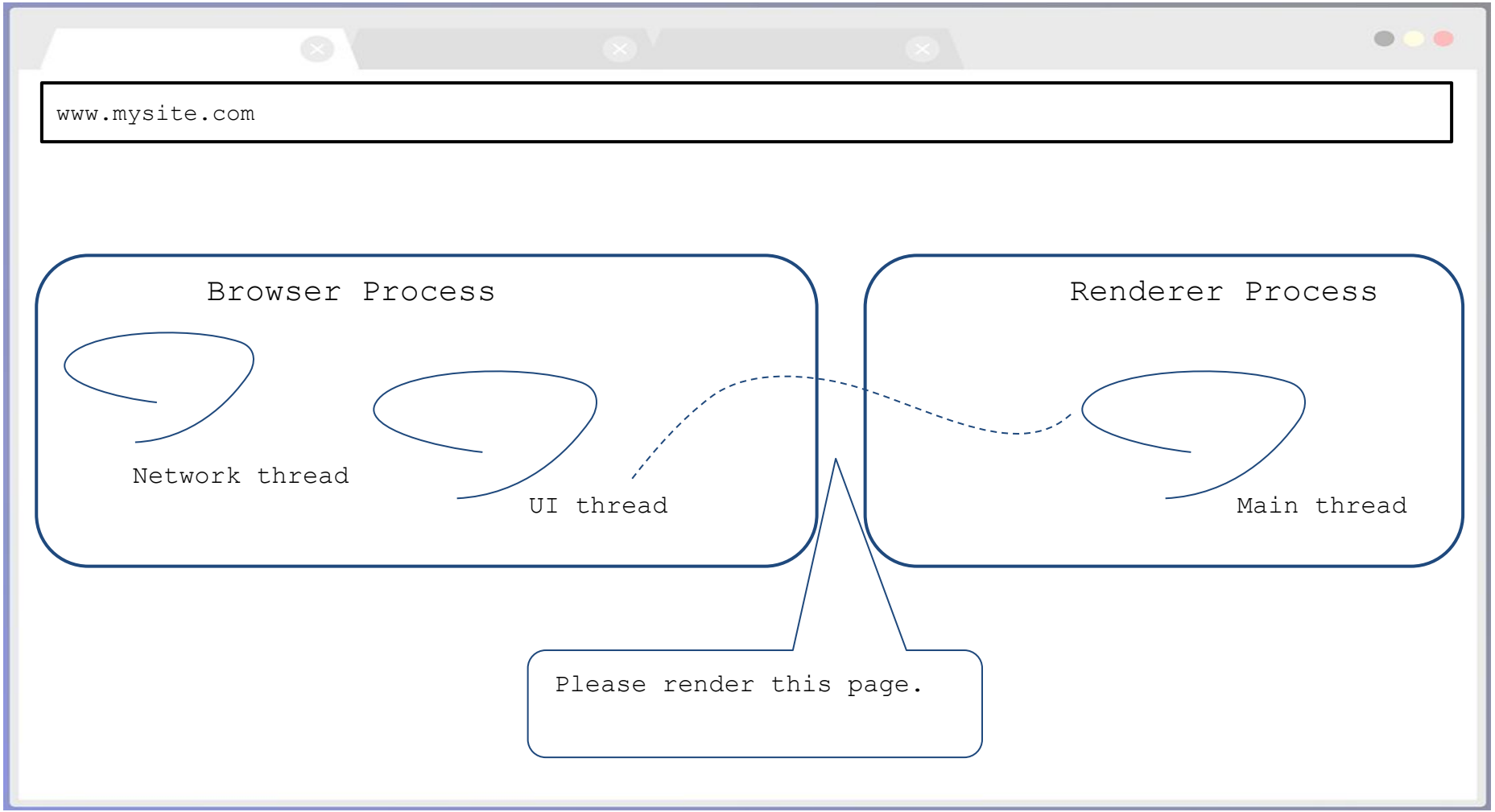
Network thread

Let me get a Rendered Process!

UI thread

Step5: Commit navigation

- An IPC is sent from browser process to renderer process to commit navigation
  - Passes data stream so renderer can receive HTML data
- Render process commits,
  - Document loading phase begins in renderer process
  - Address bar updated in browser process
    - Session history also updated so back/forward buttons work


www.mysite.com
Browser Process
Network thread
UI thread
Renderer Process
Main thread
Please render this page.

Outline

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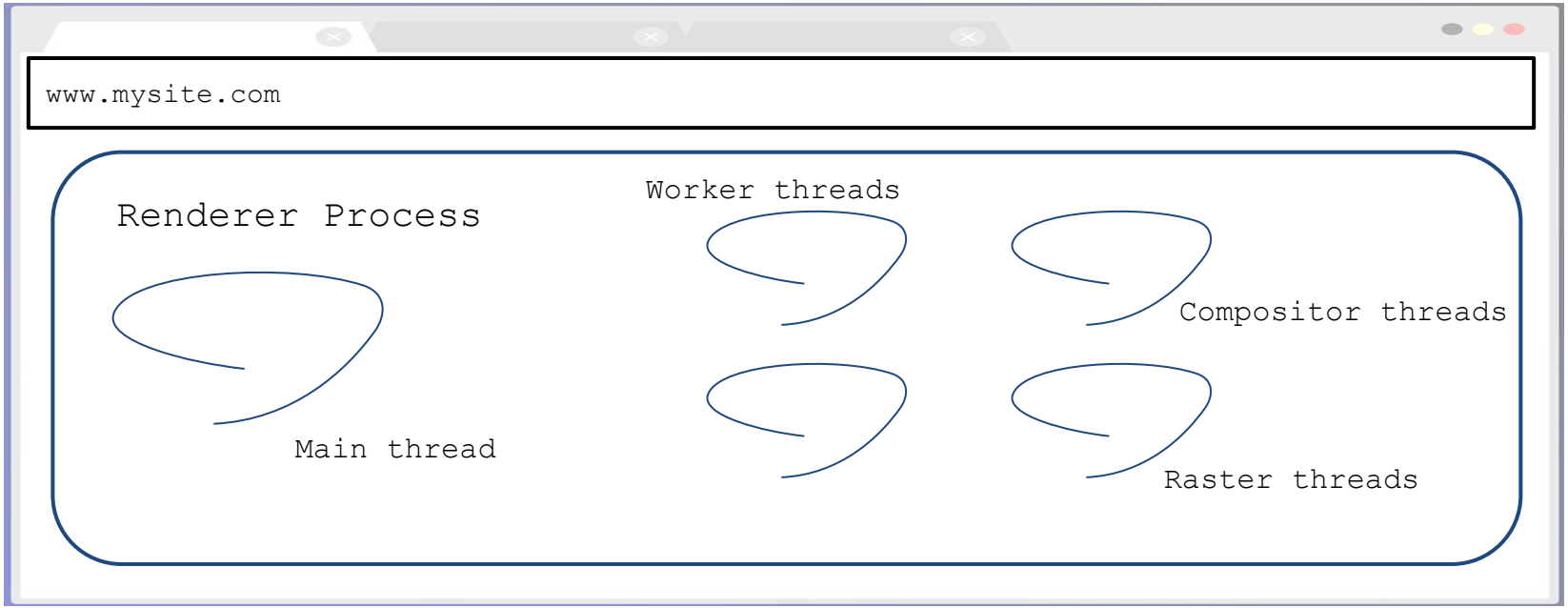
- **Internals of Renderer Process**
- Security Mechanisms within Browser

Renderer Process

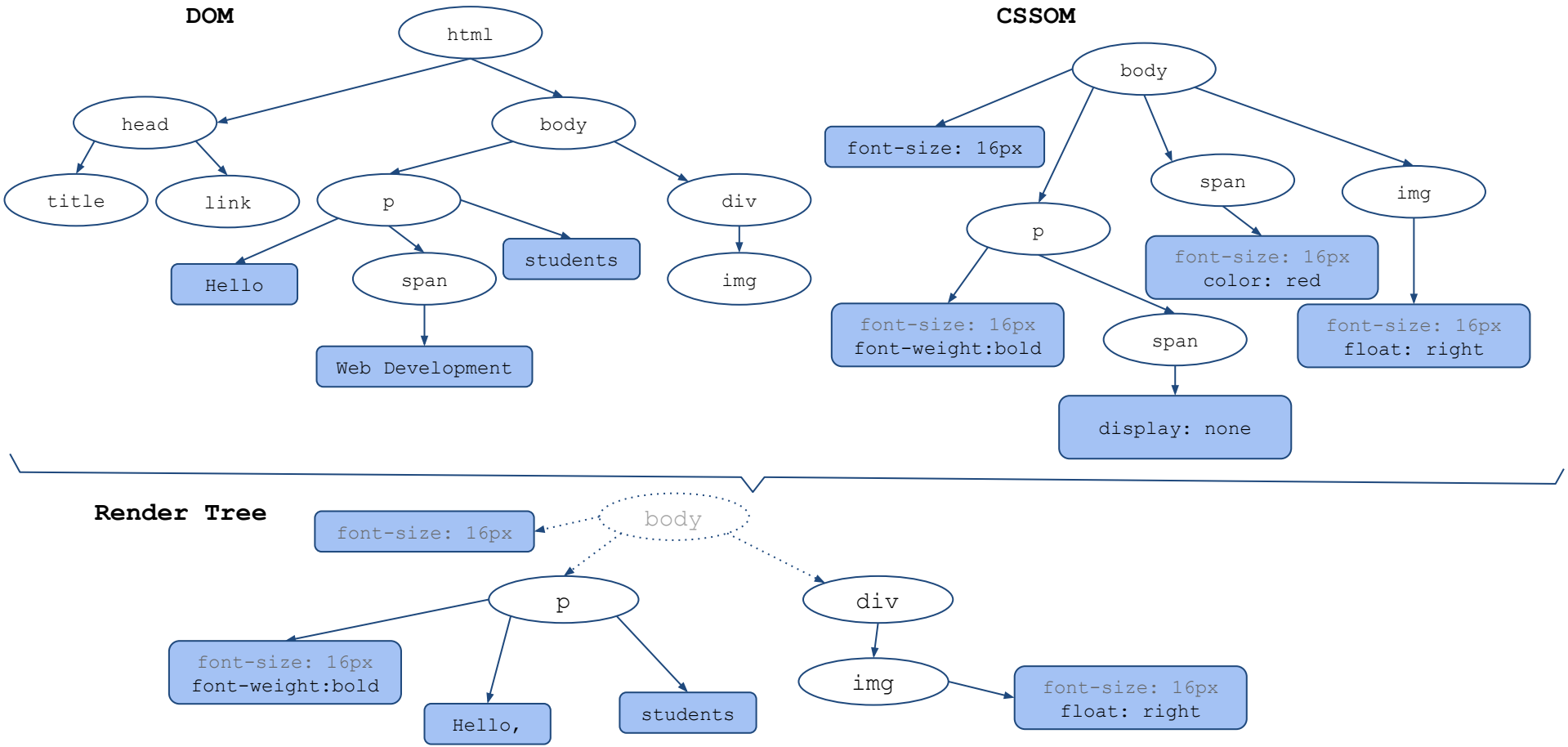
- Core job: turn HTML, CSS, and JavaScript into a web page



Steps

Step1: Parsing to construct the DOM

- Main thread parses received HTML into a Document Object Model (DOM)
  - Provides an API that web developer can use to interact via JavaScript


DOM
html
head
body
title
link
p
div
Hello
span
students
img
Web Development
CSSOM
body
font-size: 16px
p
span
img
font-size: 16px
color: red
font-size: 16px
font-weight:bold
span
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float: right
display: none
Render Tree
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Step2: Subresource loading

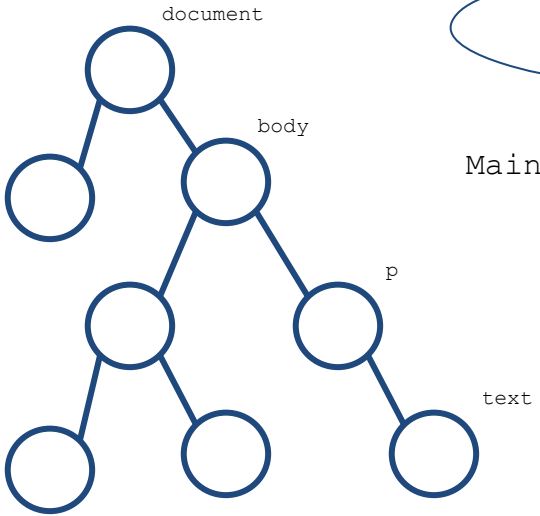
- Website usually uses external resources
  - Images, CSS, and JavaScript files
- Main thread requests them one by one while parsing
- JavaScript can block parsing till the JavaScript code is loaded and executed. Why?
  - JavaScript can change the DOM structure
  - E.g. via document.write()

www.mysite.com

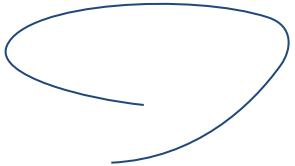
Renderer Process

Network Request

```
<!DOCTYPE html>
<html>
    <head>
        <link href="meow.css">
    </head>
    <body>
        <img src="cat.png">
        <p>
            who is cat :)?
        </p>
        <script src="meow.js">
        </script>
    </body>
</html>
```



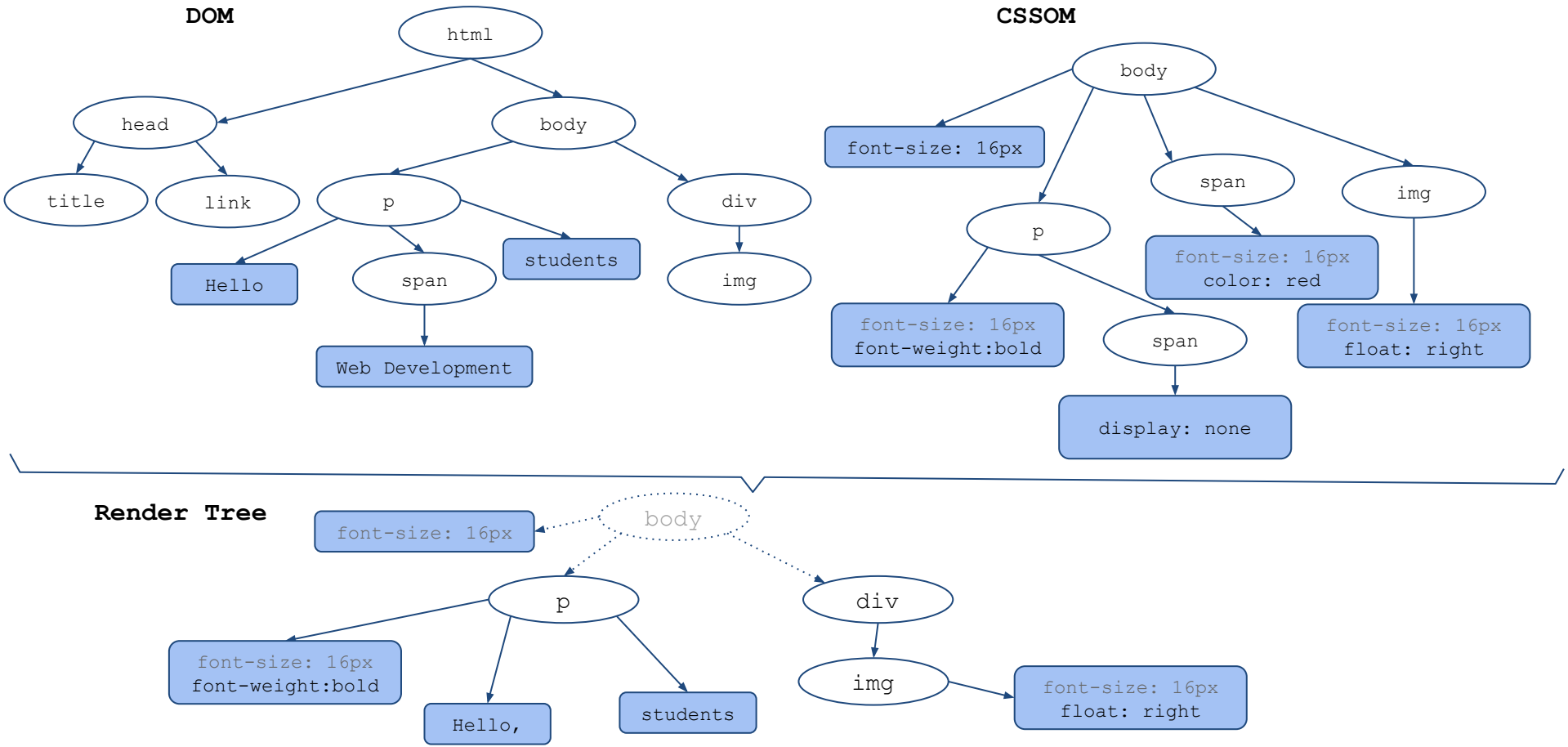
DOM Tree



Main thread

Step3: Style Calculation

- Main thread parses CSS and determines the computed style for each DOM node
  - Uses default style sheets if not provided
- The DOM and CSSOM trees combine to form the render tree
  - Render tree contains information related to what's visible on the page
    - E.g. display: none is not part of this tree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Step4: Layout

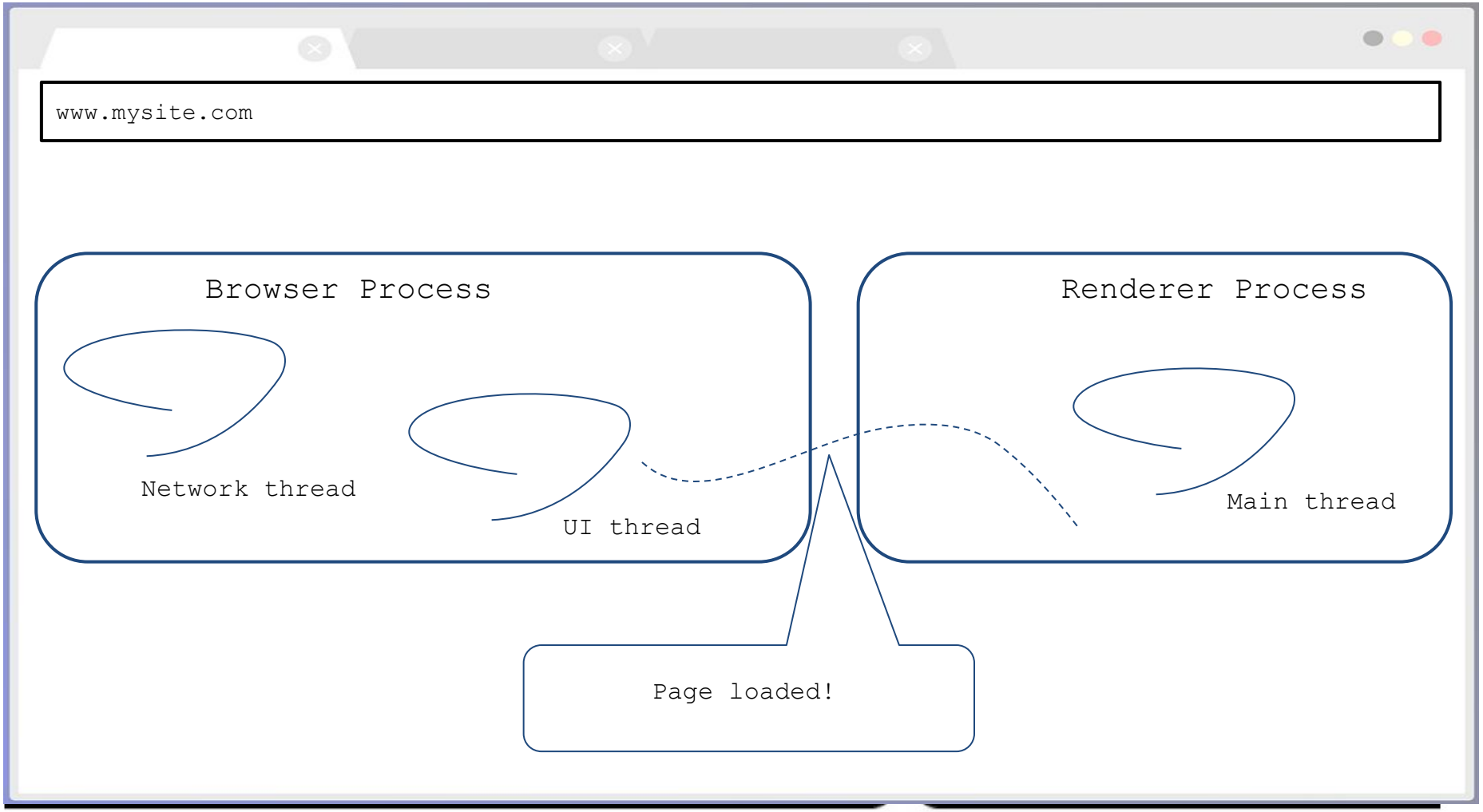
- Helps find the geometry of elements
  - Dimensions and position like x,y coordinates, bounding box sizes etc
  - Quite a challenging task
    - How big the font should be? Where to line break?

Step5: Painting and Composition

- Painting: size, shape, and location of elements known, need to paint them. Decides in what order.
- Composition: Turns above information into pixels on the screen

Post Rendering

- After rendering, render process sends an IPC to browser process
  - UI thread stops loading spinner


www.mysite.com
Browser Process
Network thread
UI thread
Renderer Process
Main thread
Page loaded!

Outline

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- **Basic Security Mechanisms within Browser**

Security Mechanisms

- Browsers implement many mechanisms to protect users from various threats
  - Many will be covered over time as we cover attacks and corresponding defenses!
- Some basic mechanisms important to know:
  - Sandboxing
  - Same Origin Policy
  - HTTPS (SSL/TLS) – covered under protocols

SandBoxing

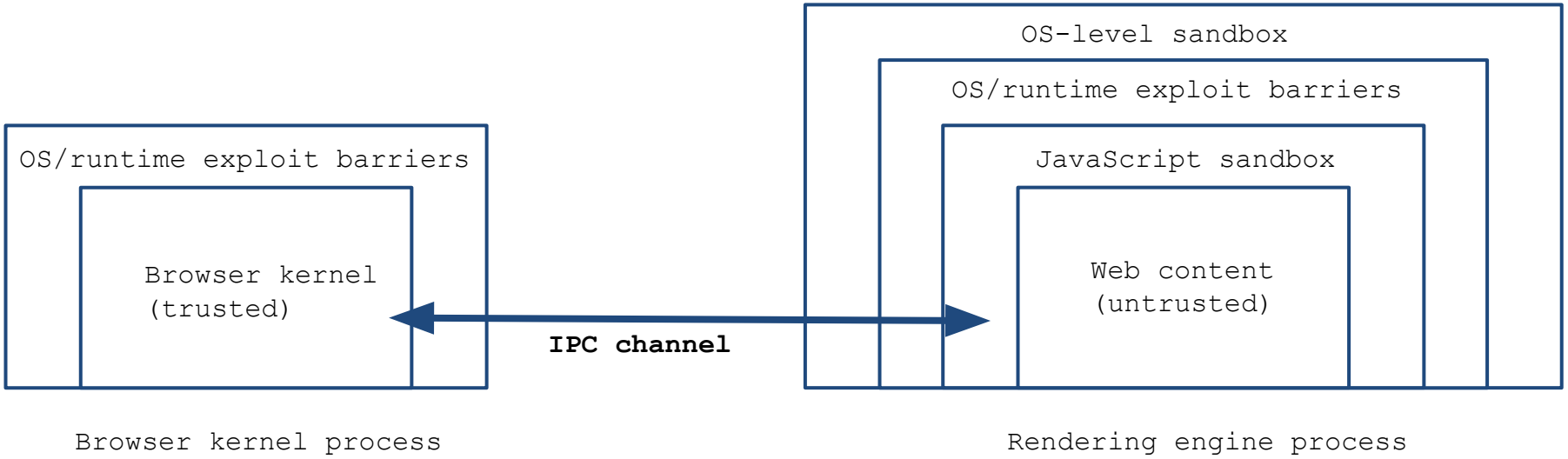
- What is a sandbox?
  - A safe and isolated environment for executing untrusted programs
    - Prevents a bad process from compromising the system
  - Achieved by restricting a process from accessing system resources
    - E.g. limit access to files outside of designated directory
- In Linux: chroot, Linux namespaces, seccomp etc establish a sandbox environment

Browser: Layers of defense

- Every large piece of software (including browser) contains bugs
- As users browse, websites can be malicious
- How to protect users?

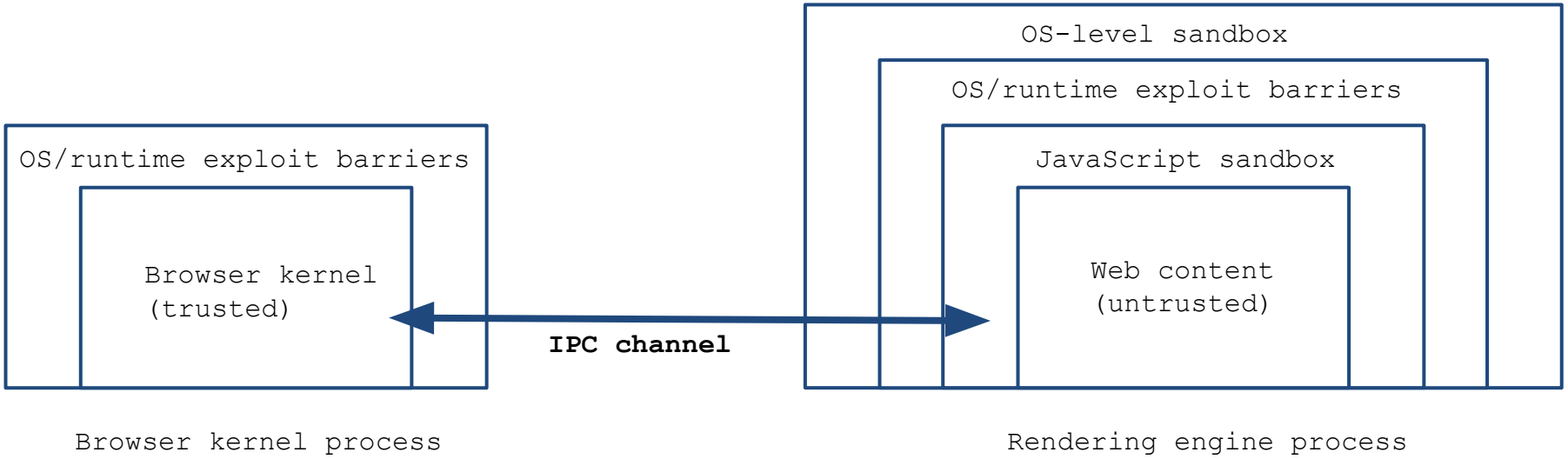
- Already saw process isolation (OS manages)!
  - Browser, Renderer (per tab), GPU etc are all different processes
- Browser process runs with high-privilege
  - Acts with user's authority
  - Responsible for UI, storing cookies (of various websites), history and network access

Layer of Defense around Google Chrome's Rendering Engine



- Rendering process runs at low privilege in a sandbox environment
  - JavaScript engine runs javascript in a sandbox
    - Prevents it from accessing sensitive resources on user's device
  - Each render process also runs in an OS-level sandbox
    - Prevents rendering process from interacting with other processes, file system etc
      - Can only exchange messages with browser process via IPC
      - Malicious code could still send messages to browser process via IPC but interface is simple and restricted to do damage!

Layer of Defense around Google Chrome's Rendering Engine



- Several OS/Runtime level protection also!
  - DEP (data execution prevention)
    - Marks memory pages (e.g. stack, heap etc) as NX (non executable)
  - ASLR (address space layout randomization)
    - Prevents attackers from guessing memory address locations
    - Can upload malicious code and jump to those locations
  - SafeSEH (safe exception handlers)
  - Stack overrun detection (GS) via canary
    - Prevents overwriting of return address
    - Prevents buffer overflow type attacks

All of above help prevent attackers from running malicious code

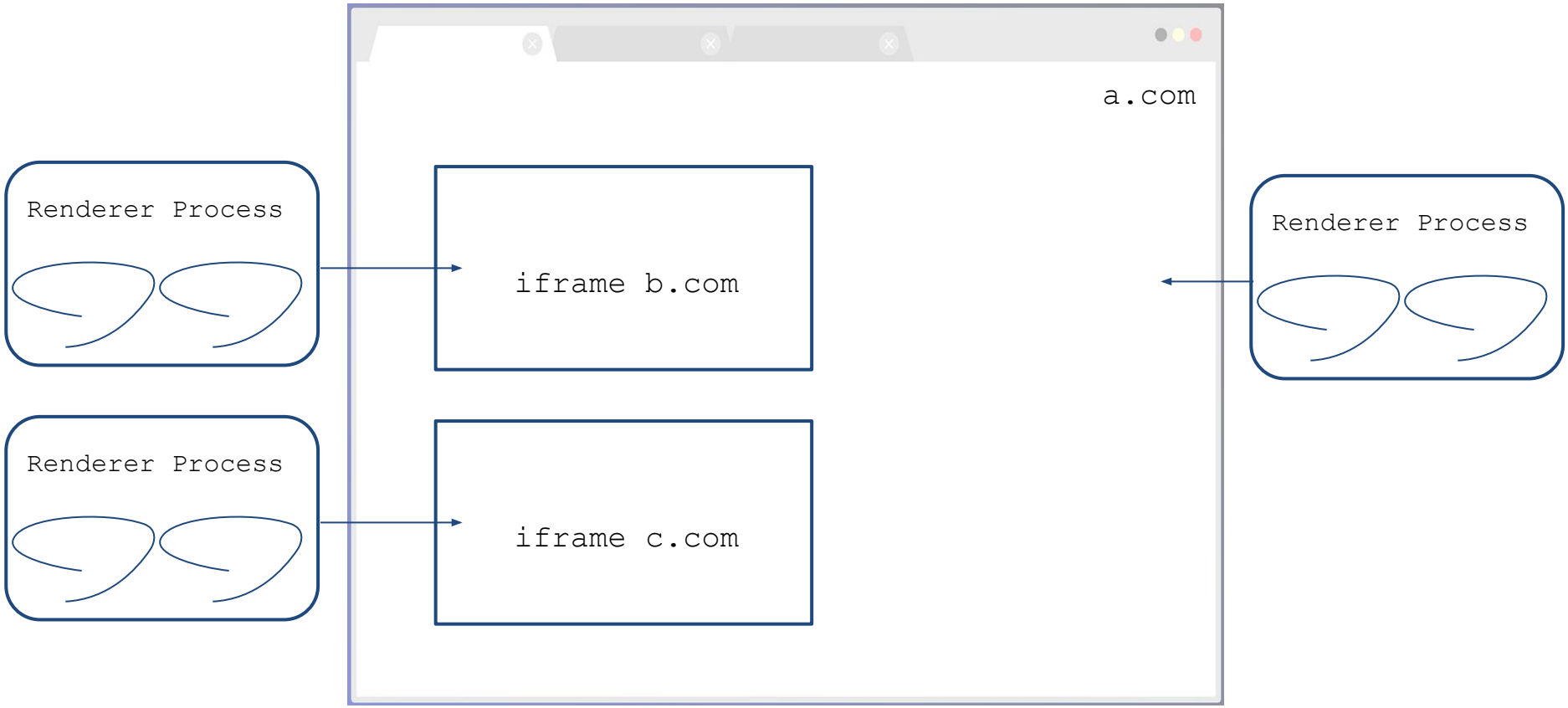
Same Origin Policy (SOP)

- SOP introduced by Netscape in 1995
  - Implemented by all browsers today
  - Result of introduction of JavaScript that manipulates DOM
- Origin: protocol + hostname + port
  - (http) + (www.iitb.ac.in) + (80)

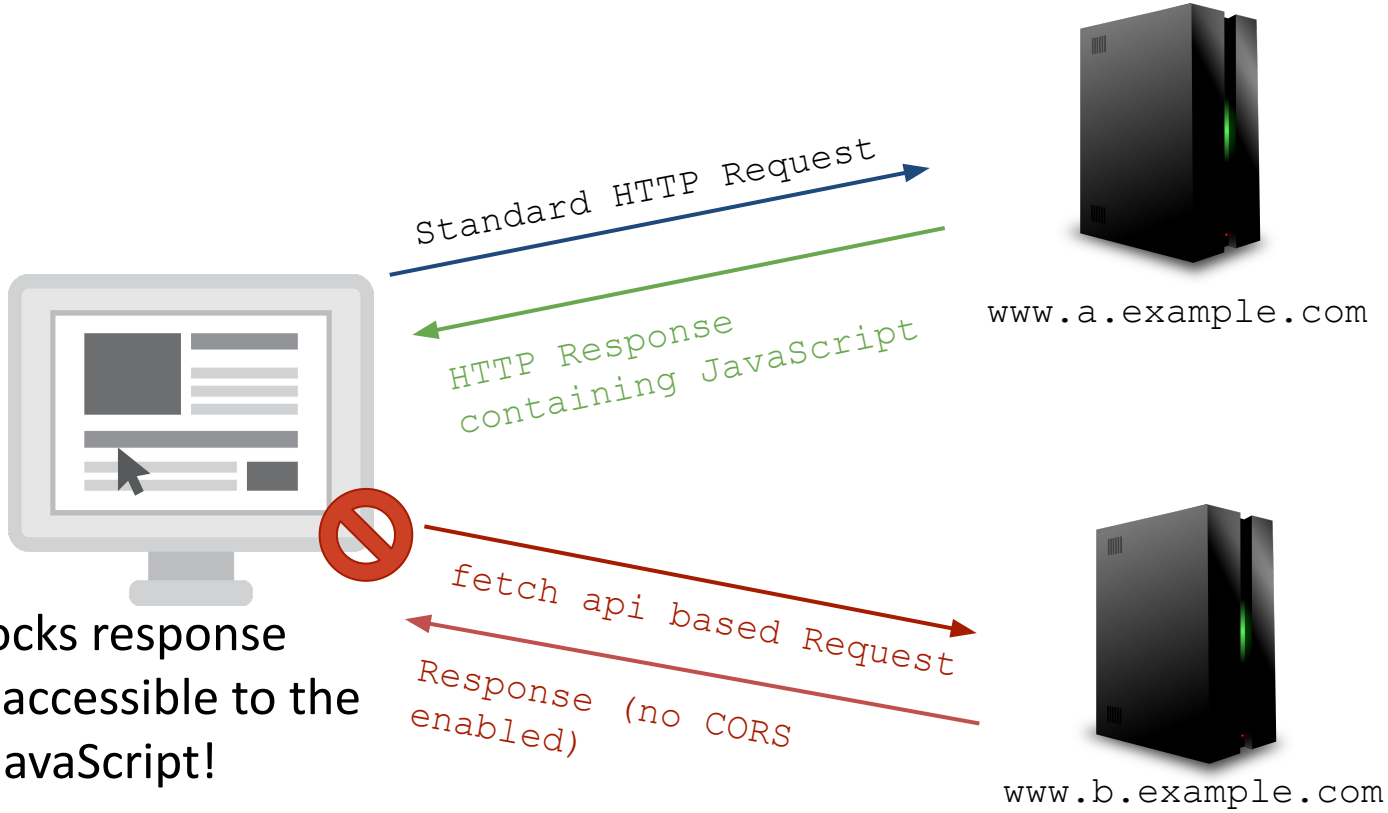
URL1	URL2	Same origin?
http://abc.org/a	http://abc.org/b	
http://abc.org	http://www.abc.org	
http://abc.org	https://abc.org	
http://abc.org:81	http://abc.org:82	

URL1	URL2	Same origin?
http://abc.org/a	http://abc.org/b	Yes
http://abc.org	http://www.abc.org	No (hostname different)
http://abc.org	https://abc.org	No (protocol different)
http://abc.org:81	http://abc.org:82	No (port different)

- SOP: Restricts how a document/script loaded by one origin can interact with a resource from another origin
  - A malicious script loaded from one origin cannot access sensitive data of another origin
  - Sensitive Data: Cookies or response to HTTP request or DOM object


a.com
Renderer Process
iframe b.com
Renderer Process
Renderer Process
iframe c.com

- Cross-origin reads via scripts are disallowed by default
  - Note: Web page can still freely embed cross-origin images, stylesheets, scripts, iframes, and videos etc
  - But “cross-origin” requests, notably Ajax requests via JavaScript are forbidden by default
    - Possible to allow via Cross-Origin Resource Sharing (CORS, will be covered later)



Browser blocks response from being accessible to the client-side JavaScript!

Many more such mechanisms

- Automatic Security Updates
- Secure Credential Management
  - Securely store and manage user credentials
- Phishing and Malware Protection
  - Browsers talk with safe browsing services that maintain databases of known phishing sites and malicious URLs
  - Warn users before navigating to such sites
- Privacy Controls
  - Include features to block or limit third-party tracking cookies
- Many policies implemented via http headers, which browsers execute (more later)

References

- https://developers.google.com/web/updates/2018/09/inside-browser-part1
- https://developer.chrome.com/blog/inside-browser-part2/
- https://developer.chrome.com/blog/inside-browser-part3/
- https://developer.mozilla.org/en-US/docs/Web/Security/Same-origin_policy
- Browser Security: Lessons from Google Chrome: https://queue.acm.org/detail.cfm?id=1556050

Summary

- Lots of action inside a browser
  - We just touched the surface! (see references for depth)
- Basic Security Mechanisms employed by browsers
  - Sandboxing
  - Same Origin Policy (SOP)
  - HTTPs (covered next)
  - Lots more (HTTP headers covered over time)